

ID	Requirement	Implemented in	Test case(s)	Result	Defect
FR-01	Load and run a pre-trained detection model on input data	OnnxYoloDetector (constructor)	TC-01	Pass	None
FR-02	Detect and classify objects in each frame	OnnxYoloDetector.DetectFrame	TC-01, TC-03	Pass (accuracy not verified)	None
FR-03	Output bounding boxes, labels and confidence scores	DetectionResult	TC-01, TC-03	Pass	None
FR-04	Process pre-recorded input as a simulated feed	DetectionPipelineRunner.ProcessBatch	TC-04, TC-11	Pass	None
FR-05	Log each frame's results and timestamp	MetricsLogger, FrameResult, SaveToJson	TC-04, TC-05	Pass	None
FR-06	Configurable simulated resource constraint before a run	ResourceConstraintProfile, ResourceConstraintSimulator, GUI profile picker	TC-06, TC-07	Partial (no memory cap)	None

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FR-07	Summary report of accuracy and performance metrics	SummaryReportGenerator, GUI results panel	TC-08, TC-09, TC-10	Partial (no accuracy metric)	DEF
FR-08	Select between at least two load conditions	LoadCondition, ProcessBatch, GUI load picker	TC-11	Pass	Non
FR-09	Stop and log an error if the model cannot load	OnnxYoloDetector (constructor)	TC-12	Pass	Non
FR-10	Skip a corrupted frame, log it, continue	DetectionPipelineRunner.ProcessFrames	TC-14	Pass	Non
FR-11	Reject a run before execution if input is missing or invalid	DetectionPipelineRunner.ValidateInputPaths	TC-13, TC-15	Pass	Non
NFR-01	Frame processing time under 100ms	SummaryReportGenerator (NFR-01 check)	TC-06, TC-08	Pass for Baseline/Baseline (87.84ms); exceeded in the other three configurations	Non

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NFR-02	At least 95% of frames sustain 15 FPS	SummaryReportGenerator (NFR-02 check)	TC-09	Fail (0.0% of frames; about 11 FPS average)	DEFECT
NFR-03	Peak memory under 512MB	FrameResult.MemoryUsedBytes, SummaryReportGenerator (NFR-03 check)	TC-08, TC-10 (see note)	Pass (300 to 322MB across configurations)	DEFECT (closed)
NFR-04	100% of frames complete under high load	SummaryReportGenerator (NFR-04 check), Parallel.For in ProcessBatch	TC-11	Pass (2/2 completed in all four configurations)	None
NFR-05	Configure and launch a run in 3 steps or fewer, without editing source	MainPage.xaml (two pickers and a run button)	Manual GUI test (Task 7)	Not automated	None
NFR-06	Results within 2% variance across 5 repeated runs	DetectionTests variance check	TC-02	Pass (0% variance)	None
NFR-07	Full run completes under at least 2 simulated hardware profiles	ResourceConstraintSimulator, Baseline and Constrained profiles	TC-07	Pass	None

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NFR-08	All defined error conditions handled without crashing	ProcessFrames try/catch, ValidateInputPaths, detector constructor, NFR-08 report check	TC-12, TC-13, TC-14, TC-15	Pass	DEF (clo DEF
NFR-09	Cold-start latency under 500ms	SummaryReportGenerator (NFR-09 check)	TC-10	Pass	Non